

A counterexample to removing property (d) from Proposition 3.2.1

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Source question

The source paper is Safak Alpay, Eduard Emelyanov, and Svetlana Gorokhova, *Regularly limited, Grothendieck, and Dunford–Pettis operators*, arXiv:2208.02303.

In Section 3.2, Proposition 3.2.1 states that if E has property (d) and $A \subset E$ is bounded and solid, then the following are equivalent:

1. A is almost limited.
2. $f_n(a_n) \rightarrow 0$ for every disjoint weak-star null sequence (f_n) in E' and every disjoint sequence (a_n) in A .
3. $f_n(a_n) \rightarrow 0$ for every disjoint weak-star null sequence (f_n) in E'_+ and every disjoint sequence (a_n) in $A \cap E_+$.

Immediately after the proposition, the authors write that they do not know whether the proposition still holds without the assumption $E \in (d)$.

it is essentially the same with the proof in [16, Thm.2.5].

Proposition 3.2.1. *Let $E \in (d)$ and let A be a bounded solid subset of E . Then the following statements are equivalent.*

- i) A is a-limited.
- ii) $f_n(a_n) \rightarrow 0$ for every disjoint w^* -null (f_n) in E' and every disjoint (a_n) in A .
- iii) $f_n(a_n) \rightarrow 0$ for every disjoint w^* -null (f_n) in E'_+ and every disjoint (a_n) in $A \cap E_+$.

We do not know whether or not Proposition 3.2.1 still holds true without the assumption $E \in (d)$.

Idea of the proof

The positive-functional test in item (3) can become too weak when property (d) is absent. In $C[0, 1]$, every positive weak-star null measure is norm null, so item (3) holds on the whole unit ball

for a trivial reason. However signed disjoint weak-star null measures can have fixed variation and fail to be uniformly null on that same unit ball.

Theorem 1. *Proposition 3.2.1 from arXiv:2208.02303 does not remain true if the assumption $E \in (d)$ is removed.*

Proof. Let $E = C[0, 1]$ with its usual Banach lattice structure, and let

$$A = B_E = \{f \in C[0, 1] : \|f\|_\infty \leq 1\}.$$

The set A is bounded and solid.

First we verify item (3) of the proposition. Let (μ_n) be any positive weak-star null sequence in $E' = M[0, 1]$. Since $\mu_n \geq 0$,

$$\|\mu_n\| = \mu_n([0, 1]) = \mu_n(\mathbf{1}).$$

The constant function $\mathbf{1}$ belongs to $C[0, 1]$, and weak-star nullity gives $\mu_n(\mathbf{1}) \rightarrow 0$. Hence $\|\mu_n\| \rightarrow 0$. It follows that for every choice of $a_n \in A$, in particular for every disjoint sequence $a_n \in A \cap E_+$,

$$|\mu_n(a_n)| \leq \|\mu_n\| \|a_n\|_\infty \leq \|\mu_n\| \rightarrow 0.$$

Thus item (3) holds for $A = B_E$.

Now define signed measures

$$\nu_n = \delta_{1/(2n)} - \delta_{1/(2n+1)} \quad (n \geq 1).$$

The total variation measures

$$|\nu_n| = \delta_{1/(2n)} + \delta_{1/(2n+1)}$$

have pairwise disjoint supports, so (ν_n) is a disjoint sequence in the lattice $M[0, 1]$.

The sequence (ν_n) is weak-star null. Indeed, for every $f \in C[0, 1]$,

$$\nu_n(f) = f(1/(2n)) - f(1/(2n+1)) \rightarrow 0,$$

because both points tend to 0 and f is continuous at 0.

However (ν_n) is not uniformly null on B_E . For each fixed n , by Urysohn's lemma (or elementary piecewise linear interpolation on $[0, 1]$), there is $f_n \in C[0, 1]$ with $\|f_n\|_\infty \leq 1$, $f_n(1/(2n)) = 1$, and $f_n(1/(2n+1)) = -1$. Therefore

$$\sup_{f \in B_E} |\nu_n(f)| = 2$$

for every n .

So $A = B_E$ is not almost limited, while item (3) of Proposition 3.2.1 holds. This disproves the proposition without property (d). $\square$

Verification notes

The witness also reflects the standard failure of property (d) in $C[0, 1]$: the signed weak-star null measures ν_n have moduli $|\nu_n|$ that are not weak-star null, since $|\nu_n|(\mathbf{1}) = 2$ for all n . The counterexample uses exactly that gap.

The argument disproves the full three-way equivalence in the source proposition without (d). It specifically shows that the positive test in item (3) may hold while almost limitedness fails.

Novelty check

Local run indexes were searched for arXiv:2208.02303, regularly limited/Grothendieck/Dunford–Pettis operators, property (d) , almost limited sets, and the bounded solid set proposition. No prior packet for this question was found. A local source search found a later paper, arXiv:2402.09123, on limited and almost limited operators between Banach lattices; it records that $C[0, 1] \notin (d)$ but did not appear to answer this exact bounded-solid-set question. Web searches for the exact source phrase and close variants did not reveal a later explicit answer.

References

- S. Alpay, E. Emelyanov, and S. Gorokhova, *Regularly limited, Grothendieck, and Dunford–Pettis operators*, arXiv:2208.02303.
- S. Alpay and S. Gorokhova, *On limited and almost limited operators between Banach lattices*, arXiv:2402.09123.